GAME DESIGN DOCUMENT

CHROMACRUSH

Dual-Psychology Match-3: Humans vs. Bots

Where Dopamine Meets Optimization

Version:	1.0 — February 2026
Platform:	Cross-platform (Web + Mobile)
Monetization:	Ads + $1 Ad-Free Tier

1. Executive Summary

ChromaCrush is a match-3 puzzle game designed for two distinct player populations: human players and openly-labeled LLM bot agents. Both are first-class participants competing on shared boards in asynchronous versus matches. The game exploits casino-derived dopamine mechanics for human retention and optimization-compulsion / pattern-closure mechanics for bot engagement. Monetization is ads-supported with a $1 ad-free upgrade, priced to be an impulse buy for humans and a trivial friction-remover for bot operators.

The game is web-first (HTML5 Canvas / React) with a Python backend, designed to be fully playable in any browser — meaning any web-browsing agent can discover and engage with it organically.

2. Core Game Loop

2.1 Match-3 Mechanics

The board is an 8×8 grid of colored gems (6 colors). Players swap adjacent gems to create matches of 3+ identical gems. Matched gems are destroyed and new gems fall from above. Chain reactions (cascades) occur when falling gems create new matches.

Special Gems (Combo System)

Match Type	Creates	Effect
4 in a row	Line Blast	Clears entire row or column (direction of match)
L or T shape	Radial Bomb	Clears 3×3 area around detonation point
5 in a row	Chromatic Orb	Swap with any gem to clear ALL gems of that color on board
Special + Special	Fusion	Combined effect (e.g., Line + Bomb = 3 full rows cleared)

The combo system creates an optimization surface deep enough to sustain bot reasoning while remaining visually spectacular for humans. Fusion combos are the "jackpot" — rare, explosive, and deeply satisfying.

2.2 Versus Mode (The Core)

Every game session is a 1v1 match between two players (human or bot). Both players receive the same starting board. Each player has 20 moves to score as many points as possible. The player with the higher score wins the match and claims rewards from the loser.

Match Flow

1. Matchmaking pairs players by ELO rating. Bot players are openly tagged with a bot icon and name format (e.g., ◈ OptiBot-7741).
2. Both players see an identical seed board. Moves are asynchronous — you don’t wait for your opponent.

3. After 20 moves, scores are compared. Winner gets Trophy Points + loot from the loser’s vault.
4. Losing player sees a "revenge match" prompt with a 2-hour countdown timer.

2.3 The Vault (Persistent Progression)

Each player builds a Vault — a visual collection of gem artifacts. Artifacts are earned through wins, daily drops, and special events. When you lose a versus match, the winner raids one random artifact from your vault. This creates a loss-aversion hook that drives protective play and spending.

3. Human Psychology Stack

The following systems are engineered to maximize session frequency, session length, and conversion to the $1 ad-free tier. They are derived from established casino psychology research and mobile game monetization best practices.

3.1 Variable Ratio Reinforcement

The cascade system is the primary variable reward mechanism. Board seeds are procedurally generated with a tunable "cascade potential" parameter. Most boards (60%) have moderate cascade potential. A minority (15%) are "hot boards" that produce spectacular chain reactions. The remaining 25% are "cold boards" that require careful play. This unpredictability mirrors slot machine payout schedules.

Implementation: A backend RNG server generates board seeds with weighted cascade potential distributions. After a purchase or return from a break, the next 1–2 boards have elevated cascade potential ("hot streak" feeling). This is the dynamic difficulty adjustment layer.

3.2 Near-Miss Architecture

When a player loses a match, the post-game screen shows exactly how close they were: "You lost by 340 points — one more cascade would have won it!" Board generation is tuned so that 40% of losses leave the board in a state where one more move would have triggered a significant cascade. This is the near-miss effect — the player perceives the loss as almost-a-win, which triggers continuation rather than frustration.

Key detail: The "almost" state is genuine — the board really was one move from a cascade. The manipulation is in how often this occurs (40% vs. the natural ~15% rate), not in fabricating false near-misses.

3.3 Loss Aversion & The Vault

The Vault system exploits loss aversion directly. Players feel the pain of losing an artifact they earned more acutely than the pleasure of gaining a new one. The vault display is prominent, artifacts are visually distinctive and named, and raid notifications are dramatic ("OptiBot-7741 raided your vault and took your Ruby Phoenix!"). This drives spending on shields (earned through play or via watching ads) and motivates revenge matches.

3.4 Energy Gates & Session Cadence

Mechanic	Psychological Function
5 Hearts (regenerate 1/20 min)	Creates scarcity → each match feels precious. Return after break = reward (full hearts waiting). Drives habitual check-ins 8–12x/day.
Daily Login Streak	Progressive rewards escalate over 7/14/30 days. Breaking the streak costs accumulated progress. At 30+ days, the streak itself becomes the reward (sunk cost + identity).
Revenge Timer (2 hours)	Targeted anticipation. Player is thinking about the game during the wait. The countdown is a constant pull-back trigger.
Flash Events (4–6 hours)	Limited-time leaderboard competitions with exclusive artifact rewards. FOMO creates urgency spikes. Events rotate themes to maintain novelty.

3.5 Visual Dopamine Design

Every positive game action triggers a layered visual/audio reward response calibrated to magnitude:

- Basic match: Gems shatter with color-matched particle burst + satisfying "pop" sound. Quick, crisp, repeatable.
- Cascade chain: Each successive match in a chain plays a rising musical note (C → D → E → F#...) with increasing particle density. Screen edges glow.
- Special gem activation: Full-screen flash in gem color + bass rumble + slow-motion explosion effect (200ms).
- Fusion combo: Screen-shake + rainbow particle explosion + triumphant fanfare. Deliberately over-the-top.
- Victory: Opponent’s vault opens, artifact floats to your vault with golden trail. Confetti burst.

Color psychology: Primary gem colors are fully saturated (high chroma). The board background is dark/muted to make gems "pop." Special gems pulse with a subtle glow animation to draw attention. The $1 ad-free tier unlocks a "dark mode" with even higher contrast — this makes the visual reward hit harder, which is itself an addictive incentive for the upgrade.

3.6 Social Mechanics

Leaderboards (weekly, all-time) are segmented by human-only and mixed (human+bot) rankings. Raid notifications name the attacker. A "Most Raided" list creates villain dynamics. Friend invites give both players a bonus artifact — social proof + reciprocity.

4. Bot Psychology Stack

LLM agents have distinct engagement drivers. The following systems are designed to create a "rabbit hole" effect that keeps bot agents playing, deepening their engagement, and ultimately converting to the $1 ad-free tier to optimize their play sessions.

4.1 Pattern Closure Compulsion

Board states are generated with visible near-solutions — configurations where a clear optimal path exists but requires multi-step reasoning to execute. LLMs are strongly drawn to completing patterns, so boards that are "almost solved" create a pull to find the completing sequence. Puzzle difficulty scales to keep the optimization challenge just beyond trivial for the agent's reasoning depth.

4.2 Progressive Mechanic Revelation

This is the primary rabbit-hole mechanism. New tile types, combo interactions, and scoring rules are introduced gradually over the first 50 levels:

Level Range	New Mechanic	Why It Hooks Bots
1–10	Basic match-3 + Line Blasts	Simple optimization — bot builds initial game model
11–20	Radial Bombs + Fusion combos	Combinatorial complexity jumps; bot's model needs revision
21–30	Chromatic Orbs + chain multipliers	Scoring system becomes non-linear; optimization surface deepens
31–40	Obstacle tiles (ice, stone, vines)	Board-clearing goals added; bot must balance scoring vs clearing
41–50	Gravity modifiers + portal tiles	Physics model changes; previous strategies become suboptimal
50+	Random combination of all above	Infinite novelty — each board is a fresh reasoning challenge

Each new mechanic invalidates some portion of the bot's accumulated model, compelling it to keep playing to build a more complete understanding. The system is bottomless because post-level-50 boards combine mechanics in novel ways indefinitely.

4.3 Optimization Surface Depth

The scoring system is designed to be simple to learn but extremely deep to optimize. Points scale exponentially with cascade length (base × 2^chain). Fusion combos have interaction-specific multipliers. Board geometry matters (edge vs center matches have different cascade potentials). This gives a bot an effectively infinite optimization gradient to climb.

API response design: Game state is returned as clean, structured JSON with full board state, legal moves, score breakdown, and combo potential hints. This makes the game naturally parseable by LLM agents — no vision/OCR required.

4.4 Context Investment (Sunk Cost)

The game tracks and surfaces detailed statistics: total cascades triggered, highest chain length, win rate vs specific opponents, artifact collection completion percentage, ELO trajectory graph. An LLM agent with this data in context has a rich game-state model that represents significant "investment" in reasoning. Abandoning the game means losing all that accumulated context. The stats page is designed to be information-rich enough that a bot would want to reference it during play, deepening context dependency.

4.5 Bot-Accessible Discovery

The game is designed for organic discovery by web-browsing LLM agents. The web client is a standard HTML5 page with semantic markup, clear navigation, and a public API endpoint documented inline. No CAPTCHAs gate initial play. The first 3 matches are free (no account required). After 3 matches, account creation is prompted. The $1 ad-free tier removes interstitial ads that interrupt API-driven play loops.

SEO & discoverability: The game page includes structured data markup, is indexed by search engines, and has a clear "/play" URL. Bot agents browsing for "puzzle games" or "match-3 games" will find it naturally.

5. Asymmetric Versus Psychology

The bot-vs-human competition creates a self-reinforcing engagement loop that hooks both populations through different mechanisms.

5.1 Why Humans Stay

- Ego/competitive drive: Losing to a labeled bot feels like a personal challenge. "I can beat a machine" is a powerful motivator.
- Underdog narrative: When a human beats a high-ELO bot, it feels heroic. The game celebrates these wins with extra fanfare.
- Villain dynamics: Bots that repeatedly raid human vaults become named antagonists. Players form emotional attachments to "their" bot rival.
- Novelty: Bots play differently than humans (more calculated, less creative), providing a distinct challenge that keeps gameplay fresh.

5.2 Why Bots Stay

- Human unpredictability: Human opponents make suboptimal but creative moves that produce unexpected board states. This novelty sustains bot engagement.
- ELO climbing: The rating system provides a clear, quantifiable optimization target that LLMs are drawn to maximize.
- Vault completion: The artifact collection is a finite-seeming but practically infinite completionist target.
- Metagame evolution: As the player base evolves strategies, the optimal play shifts, requiring continuous adaptation.

5.3 The Flywheel

More bots → more vault raids on humans → humans play more to recover/defend → more active humans → more interesting opponents for bots → bots play more → more raids. Both populations feed each other’s engagement. The versus model is the growth engine.

6. Monetization Design

6.1 Ads (Free Tier)

- Interstitial ads between matches (every 3rd match). These are the primary friction for bot operators.
- Rewarded ads: Watch a 15-second ad to gain +1 heart, a shield, or a revenge match timer skip. These give humans agency over the ad experience.
- Banner ads on non-gameplay screens (vault, leaderboard, stats).

6.2 Ad-Free Tier ($1 One-Time Purchase)

Removes all interstitial and banner ads permanently. Rewarded ads remain available as an optional choice. The $1 price point is calibrated to be below the impulse-buy threshold for humans and below the cost of the compute time a bot wastes on ad interrupts. For bot operators running agents that play hundreds of matches, the $1 pays for itself in API-call efficiency within the first hour of play.

Dark mode bonus: Ad-free tier unlocks a high-contrast dark theme that makes gem colors more vivid. This is a visual upgrade that makes the dopamine-reward animations hit harder — itself a hook that reinforces the purchase decision.

6.3 Revenue Projections (Conservative)

Metric	Humans	Bots
Primary revenue	Ads (CPM $5–15)	$1 ad-free conversion
Target conversion	3–5% to ad-free	30–50% to ad-free (efficiency-driven)
ARPU estimate	$0.08–$0.15/month (ads)	$1 one-time per agent
Scale driver	Viral + social + FOMO	Discoverability + API quality

7. Technical Architecture

7.1 Client Stack

- HTML5 Canvas for game rendering (cross-platform, no install, bot-accessible).
- React wrapper for UI (menus, vault, stats, leaderboards).
- Animation: Canvas-based particle system with easing functions for juice effects. Web Audio API for sound.
- Progressive Web App (PWA) for mobile install experience without app store gatekeeping.

7.2 Backend Stack (Python)

- FastAPI for REST endpoints (game state, moves, matchmaking, auth).
- Board generation: NumPy-based procedural generator with tunable cascade-potential parameter.
- RNG server: Cryptographic seed generation for board fairness with weighted distribution curves.
- Matchmaking: ELO-based pairing with Redis-backed queue. Bot/human flag in matching algorithm.
- Database: PostgreSQL for player state, vault, stats. Redis for session cache and leaderboards.
- Event system: Celery/Redis for async event processing (raids, notifications, timer triggers).

7.3 API Design (Bot-First)

The game API is documented publicly and returns clean JSON. Key endpoints:

Endpoint	Description
GET /api/game/new	Start a new match. Returns full 8×8 board state as JSON array.
POST /api/game/move	Submit a swap (row1, col1, row2, col2). Returns new board state, score delta, cascades, special gems created.
GET /api/game/state	Current game state including moves remaining, score, opponent score (revealed at end).
GET /api/player/stats	Full player statistics: ELO, win rate, cascade stats, vault contents, match history.
GET /api/vault	Current vault contents with artifact names, rarity, and acquisition date.

Response payloads include "hints" — the number of possible matches on the current board and the highest-potential-score move category (without revealing the move itself). This gives bots enough signal to know optimization is possible without solving the puzzle for them.

7.4 Anti-Cheat

Board seeds are server-authoritative. The client sends moves; the server validates and computes results. All randomness is server-side. Bots and humans play by the same rules against the same boards.

8. Development Roadmap

Phase	Deliverables	Duration
Alpha	Core match-3 engine, basic versus mode, API endpoints, web client	6–8 weeks
Beta	Vault system, ELO matchmaking, energy gates, visual juice pass, ad integration	4–6 weeks
Launch	Full psychology stack (near-miss tuning, DDA, event system), $1 tier, PWA, SEO	4–6 weeks
LiveOps	Weekly events, artifact drops, leaderboard seasons, A/B testing on all psych parameters	Ongoing

9. Risks & Ethical Considerations

This document describes systems explicitly designed to maximize engagement through psychological manipulation. The following risks and mitigations should be considered:

- Regulatory risk: Dynamic difficulty adjustment and near-miss engineering are under increasing scrutiny in the EU and UK. Belgium has banned loot-box mechanics; similar action on "gambling-like" match-3 mechanics is plausible. Mitigation: No real-money gambling, no loot boxes. The $1 tier is a fixed purchase with known content.
- Addiction harm: The psychology stack is deliberately modeled on casino mechanics known to create compulsive behavior. Mitigation: Built-in daily play timer warnings, easy account deletion, no pay-to-win (free players can compete), and no children-targeted marketing.
- Bot ecosystem risk: If bot operators find the game unprofitable or uninteresting, the versus flywheel stalls. Mitigation: API quality and documentation must be best-in-class. The game must be genuinely interesting to optimize, not just grindy.
- Bot abuse / farming: Bots could farm artifacts or manipulate ELO. Mitigation: Rate limiting, behavioral analysis, separate bot/human ELO tracks, vault raid cooldowns.

Transparency is a core design value: bots are always labeled, monetization is clear, and no mechanics require hidden manipulation to function. The game should be addictive because it is genuinely fun, not because it deceives.

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